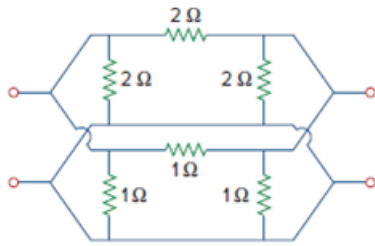


Problem

Obtain the h parameters for the network in Fig. 19.115.

Figure 19.115



Step-by-step solution

Step 1 of 13

Refer to Figure 19.115 in the textbook.

Draw the separate circuits for the two individual networks connected in parallel and find the y parameters.

The circuit for the upper network is,

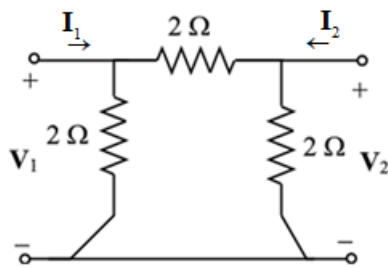


Figure 1

Step 2 of 13

Define y parameters as follows:

$$I_1 = y_{11} V_1 + y_{12} V_2$$

$$I_2 = y_{21} V_1 + y_{22} V_2$$

Therefore,

$$y_{11} = \left. \frac{I_1}{V_1} \right|_{V_2=0}$$

$$y_{12} = \left. \frac{I_1}{V_2} \right|_{V_1=0}$$

$$y_{21} = \left. \frac{I_2}{V_1} \right|_{V_2=0}$$

$$y_{22} = \left. \frac{I_2}{V_2} \right|_{V_1=0}$$

Step 3 of 13

Apply current source $\mathbf{I}_1$ and leave the output port short to determine $\mathbf{y}_{11}$ and $\mathbf{y}_{21}$.

The $2\ \Omega$ resistor in the output side is shorted due to short-circuit across it.

Apply Kirchhoff's current law at the node $\mathbf{V}_1$.

$$\begin{aligned}\mathbf{I}_1 &= \frac{\mathbf{V}_1}{2} + \frac{\mathbf{V}_1 - \mathbf{V}_2}{2} \\ &= \frac{\mathbf{V}_1}{2} + \frac{\mathbf{V}_1 - 0}{2} \quad (\text{since, } \mathbf{V}_2 = 0) \\ &= \mathbf{V}_1 \left(\frac{1}{2} + \frac{1}{2} \right) \\ &= \mathbf{V}_1\end{aligned}$$

Therefore,

$$\begin{aligned}\mathbf{y}_{11} &= \left. \frac{\mathbf{I}_1}{\mathbf{V}_1} \right|_{\mathbf{V}_2=0} \\ &= 1\end{aligned}$$

Step 4 of 13

Apply Kirchhoff's current law at the node $\mathbf{V}_2$.

$$\begin{aligned}\frac{\mathbf{V}_2 - \mathbf{V}_1}{2} + \frac{\mathbf{V}_2}{2} - \mathbf{I}_2 &= 0 \\ \mathbf{I}_2 &= \frac{0 - \mathbf{V}_1}{2} + 0 \quad (\text{since, } \mathbf{V}_2 = 0) \\ \frac{\mathbf{I}_2}{\mathbf{V}_1} &= \frac{-1}{2}\end{aligned}$$

Therefore,

$$\begin{aligned}\mathbf{y}_{21} &= \left. \frac{\mathbf{I}_2}{\mathbf{V}_1} \right|_{\mathbf{V}_2=0} \\ &= \frac{-1}{2}\end{aligned}$$

Step 5 of 13

Apply current source $\mathbf{I}_2$ and leave the input port short to determine $\mathbf{y}_{22}$ and $\mathbf{y}_{12}$.

The $2\ \Omega$ resistor in the input side is shorted due to short-circuit across it.

Apply Kirchhoff's current law at the node $\mathbf{V}_2$.

$$\begin{aligned}\mathbf{I}_2 &= \frac{\mathbf{V}_2 - \mathbf{V}_1}{2} + \frac{\mathbf{V}_2}{2} \\ &= \frac{\mathbf{V}_2 - 0}{2} + \frac{\mathbf{V}_2}{2} \quad (\text{since, } \mathbf{V}_1 = 0) \\ &= \mathbf{V}_2 \left(\frac{1}{2} + \frac{1}{2} \right) \\ &= \mathbf{V}_2\end{aligned}$$

Therefore,

$$\begin{aligned}\mathbf{y}_{22} &= \left. \frac{\mathbf{I}_2}{\mathbf{V}_2} \right|_{\mathbf{V}_1=0} \\ &= 1\end{aligned}$$

Step 6 of 13

Apply Kirchhoff's current law at the node V_1 .

$$\frac{V_1 - V_2}{2} + \frac{V_1}{2} - I_1 = 0$$

$$I_1 = \frac{0 - V_2}{2} + 0 \quad (\text{since, } V_1 = 0)$$

$$\frac{I_1}{V_2} = \frac{-1}{2}$$

Therefore,

$$y_{12} = \left. \frac{I_1}{V_2} \right|_{V_1=0}$$

$$= \frac{-1}{2}$$

Thus, the y parameters of the upper network are $\begin{bmatrix} 1 & -0.5 \\ -0.5 & 1 \end{bmatrix} \text{ S}$.

Step 7 of 13

The circuit for the lower network is,

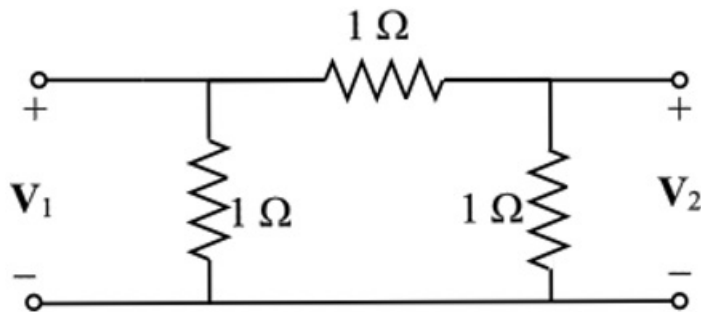


Figure 2

Step 8 of 13

Apply current source I_1 and leave the output port short to determine y_{11} and y_{21} .

The $1\ \Omega$ resistor in the output side is shorted due to short-circuit across it.

Apply Kirchhoff's current law at the node V_1 .

$$I_1 = \frac{V_1}{1} + \frac{V_1 - V_2}{1}$$

$$= V_1 + V_1 - 0 \quad (\text{since, } V_2 = 0)$$

$$= 2V_1$$

Therefore,

$$y_{11} = \left. \frac{I_1}{V_1} \right|_{V_2=0}$$

$$= 2$$

Step 9 of 13

Apply Kirchhoff's current law at the node V_2 .

$$\frac{V_2 - V_1}{1} + \frac{V_2}{1} - I_2 = 0$$

$$I_2 = 0 - V_1 + 0 \quad (\text{since, } V_2 = 0)$$

$$\frac{I_2}{V_1} = -1$$

Therefore,

$$\begin{aligned} y_{21} &= \left. \frac{I_2}{V_1} \right|_{V_2=0} \\ &= -1 \end{aligned}$$

Step 10 of 13

Apply current source I_2 and leave the input port short to determine y_{22} and y_{12} .

The $1\ \Omega$ resistor in the input side is shorted due to short-circuit across it.

Apply Kirchhoff's current law at the node V_2 .

$$\begin{aligned} I_2 &= \frac{V_2 - V_1}{1} + \frac{V_2}{1} \\ &= V_2 - 0 + V_2 \quad (\text{since, } V_1 = 0) \\ &= 2V_2 \end{aligned}$$

Therefore,

$$\begin{aligned} y_{22} &= \left. \frac{I_2}{V_2} \right|_{V_1=0} \\ &= 2 \end{aligned}$$

Step 11 of 13

Apply Kirchhoff's current law at the node V_1 .

$$\frac{V_1 - V_2}{1} + \frac{V_1}{1} - I_1 = 0$$

$$I_1 = 0 - V_2 + 0 \quad (\text{since, } V_1 = 0)$$

$$\frac{I_1}{V_2} = -1$$

Therefore,

$$\begin{aligned} y_{12} &= \left. \frac{I_1}{V_2} \right|_{V_1=0} \\ &= -1 \end{aligned}$$

Thus, the y parameters of the lower network are $\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \text{ S}$.

Step 12 of 13

Determine the y -parameters of the total network.

$$y_{11} = 1 + 2$$

$$= 3$$

$$y_{12} = -0.5 - 1$$

$$= -1.5$$

$$y_{21} = -\frac{1}{2} - 1$$

$$= -1.5$$

$$y_{22} = 1 + 2$$

$$= 3$$

Thus, the y parameters of the upper network are $\begin{bmatrix} 3 & -1.5 \\ -1.5 & 3 \end{bmatrix} \text{ S}$.

Determine Δy .

$$\Delta y = (3)(3) - (-1.5)(-1.5)$$

$$= 9 - 2.25$$

$$= 6.75$$

Step 13 of 13

Use the relationships between the parameters to determine h parameters.

$$h_{11} = \frac{1}{y_{11}}$$

$$= \frac{1}{3}$$

$$= 0.34$$

$$h_{12} = -\frac{y_{12}}{y_{11}}$$

$$= -\frac{-1.5}{3}$$

$$= 0.5$$

$$h_{21} = \frac{y_{21}}{y_{11}} \quad h_{22} = \frac{\Delta y}{y_{11}}$$

$$= \frac{-1.5}{3} \quad = \frac{6.75}{3}$$

$$= -0.5 \quad = 2.25$$

Thus, the h parameters of the network are $\begin{bmatrix} 0.34 \, \Omega & 0.5 \\ -0.5 & 2.25 \, \text{S} \end{bmatrix}$.